ASSIGNMENT 5 — QUESTION 8

Custom Array myMap()

JavaScript (ES6+) solution

```
Array.prototype.myMap = function (callback) {
    const result = [];

    for (let index = 0; index < this.length; index++) {
        if (index in this) {
            result.push(callback(this[index], index, this));
        }
    }

    return result;
};

console.log([1, 2, 3].myMap(num => num * 2));
console.log([1, 2, 3].myMap((num, index) => num + index));
```

Expected Output

```
[ 2, 4, 6 ]
[ 1, 3, 5 ]
```

Explanation

myMap() loops through the array, calls the callback with the current value, index, and original array, and stores the results in a new array. It skips empty slots in sparse arrays, like the built-in map(). In production code, avoid modifying built-in prototypes unless required.